北京交通大學

信号与系统

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State variable analysis of the system

※ Basic concepts and general forms of system state variable analysis

※ Establishment of state equation and output equation for continuous time system

※ Establishment of state equations and output equations for discrete-time systems

※ The solution of state equation and output equation of continuous time system

※ The solution of state equation and output equation of discrete-time system

Establishment of state equations and output equations for discrete-time systems

Discrete systems are often described in the form of difference equations, system functions, or simulation block diagrams, which can be established by the state equations and output equations of the discrete system.

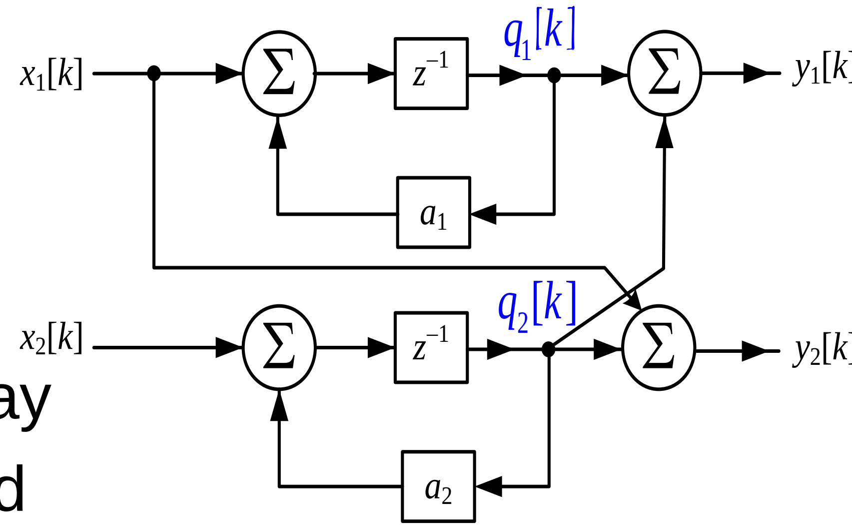
※ The state equation and output equation are established from the difference equation

※ State equation and output equation are established by simulating block diagram

※ State equations and output equations are established by system functions

State equation and output equation are established by simulating block diagram

[Example] Given to describe a discrete system simulation block diagram, establish its state equation and output equation.



Solution: The system has two delay controllers, whose output $q_1[k]$ and $q_2[k]$ are selected as state variables.

Write the equation of state around the input column of the delay device:

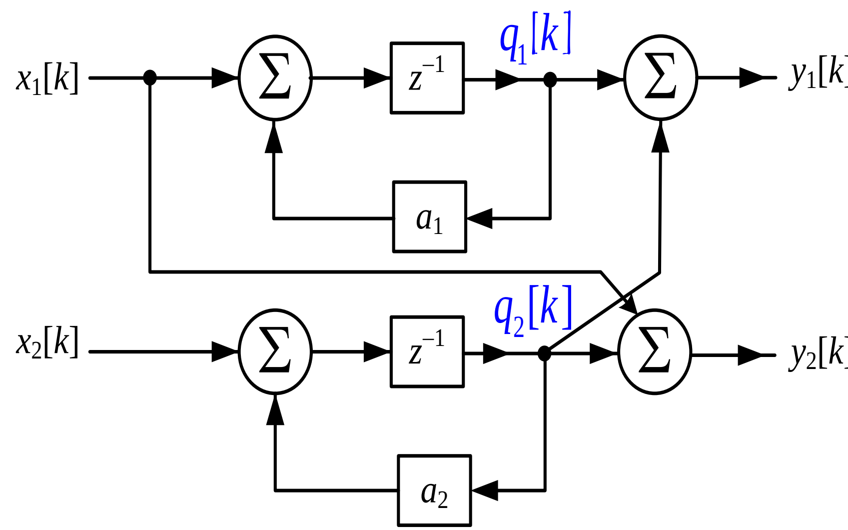
$$q_1[k+1] = a_1 q_1[k] + x_1[k] \quad q_2[k+1] = a_2 q_2[k] + x_2[k]$$

Write the output equation around the discrete system output column:

$$y_1[k] = q_1[k] + q_2[k] \qquad y_2[k] = q_2[k] + x_1[k]$$

State equation and output equation are established by simulating block diagram

[Example] Given to describe a discrete system simulation block diagram, establish its state equation and output equation



Solution: written in matrix form:

$$\begin{bmatrix} q_1[k+1] \\ q_2[k+1] \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix}$$

$$\begin{bmatrix} y_1[k] \\ y_2[k] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix}$$

State equation and output equation are established by simulating block diagram

(1) Select the output end of the delay device as the state variable;

(2) Write the equation of state around the input column of the delay device;

(3) Write the output equation around the output columns of the discrete system.

State equations and output equations are established by system functions

According to the difference equation or system function describing the discrete system, the corresponding simulation block diagram is drawn, and then the state equation and output equation of the system are established from the simulation block diagram.

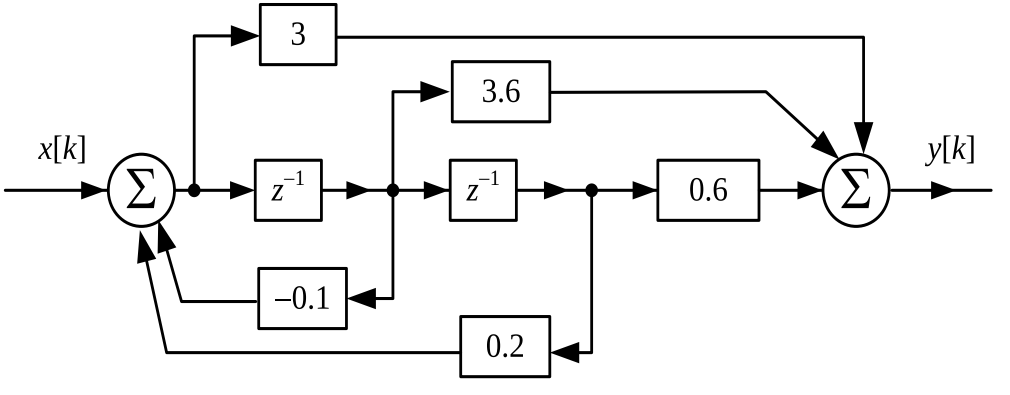
State equations and output equations are established by system functions

[Example]The system function describing a discrete system is known as

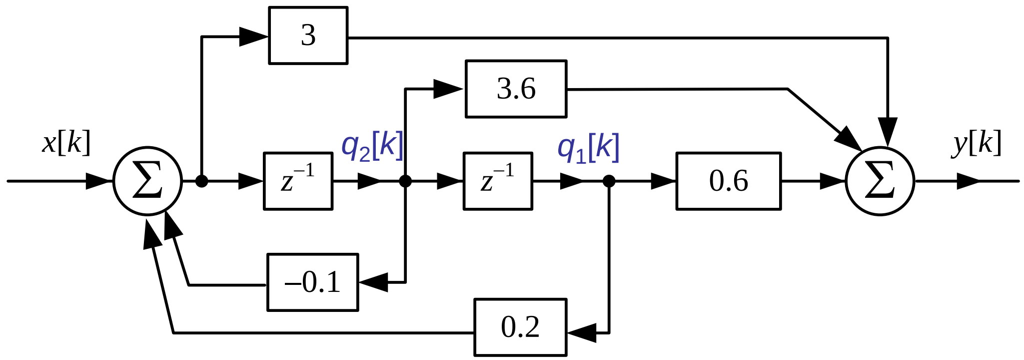
$$H(z) = \frac{3 + 3.6z^{-1} + 0.6z^{-2}}{1 + 0.1z^{-1} - 0.2z^{-2}}$$

The simulation block diagrams of direct, cascading and parallel structures of the system are drawn, and the state equations and output equations of the system are established.

Solution: (1) Direct block diagram



State equations and output equations are established by system functions



Select the output end of the delay device as the state variables $q_1[k]$ and $q_2[k]$ of the system:

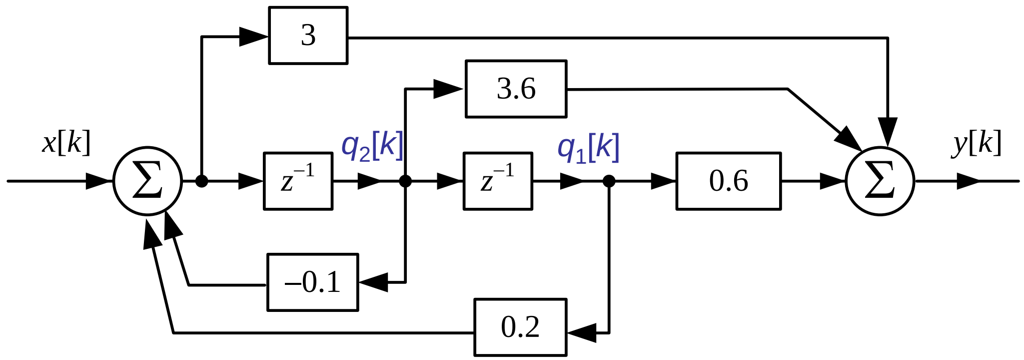
Write the equation of state around the input column of the delay device: $q_1[k+1] = q_2[k]$ $\quad q_2[k+1] = 0.2q_1[k] - 0.1q_2[k] + x[k]$

Write the output equation around the discrete system output column:

$$y[k] = 0.6q_1[k] + 3.6q_2[k] + 3(0.2q_1[k] - 0.1q_2[k] + x[k])$$

$$= 1.2q_1[k] + 3.3q_2[k] + 3x[k])$$

State equations and output equations are established by system functions



Matrix form of the equation of state:

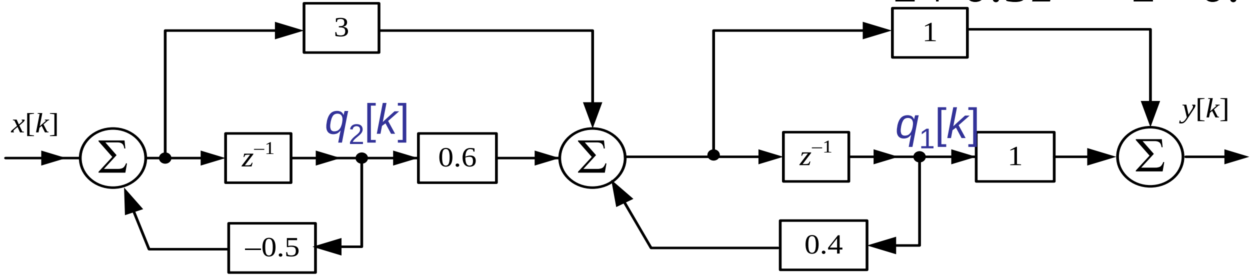
$$\begin{bmatrix} q_1[k+1] \\ q_2[k+1] \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0.2 & -0.1 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x[k]$$

Matrix form of the output equation:

$$y[k] = \begin{bmatrix} 1.2 & 3.3 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + 3x[k]$$

State equations and output equations are established by system functions

(2) Cascading block diagram $$H(z)=\frac{3+0.6z^{-1}}{1+0.5z^{-1}}\cdot\frac{1+z^{-1}}{1-0.4z^{-1}}$$



State variables $q_1[k]$ and $q_2[k]$ are selected at the output end of the delayer:

Write the equation of state from the input column of the delay device:

$$q_2[k+1] = -0.5q_2[k] + x[k]$$

$$q_1[k+1] = 0.4q_1[k] + 0.6q_2[k] + 3(-0.5q_2[k] + x[k])$$

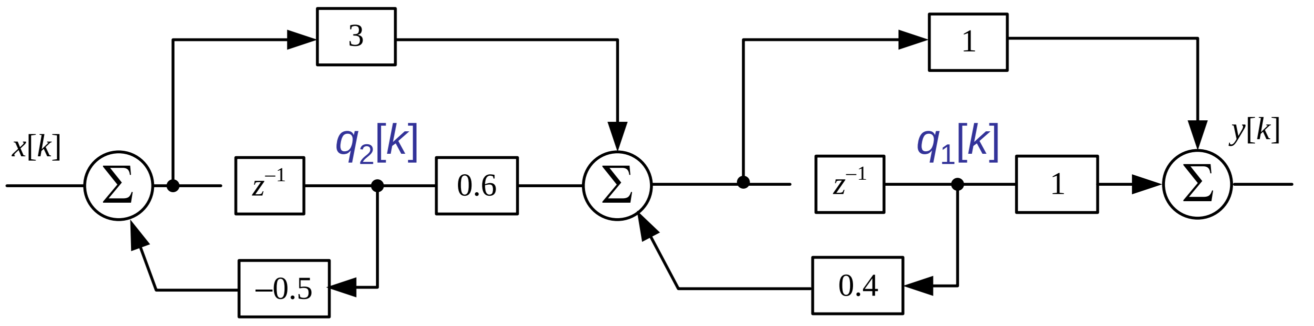
$$= 0.4q_1[k] - 0.9q_2[k] + 3x[k]$$

Write the output equation from the discrete system output column:

$$y[k] = q_1[k] + q_1[k+1] = 1.4q_1[k] - 0.9q_2[k] + 3x[k]$$

State equations and output equations are established by system functions

(2) Cascading block diagram $H(z)=\dfrac{3+0.6z^{-1}}{1+0.5z^{-1}}\cdot\dfrac{1+z^{-1}}{1-0.4z^{-1}}$



Matrix form of the equation of state:

$$\begin{bmatrix} q_1[k+1] \\ q_2[k+1] \end{bmatrix} = \begin{bmatrix} 0.4 & -0.9 \\ 0 & -0.5 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} x[k]$$

Matrix form of the equation of output:

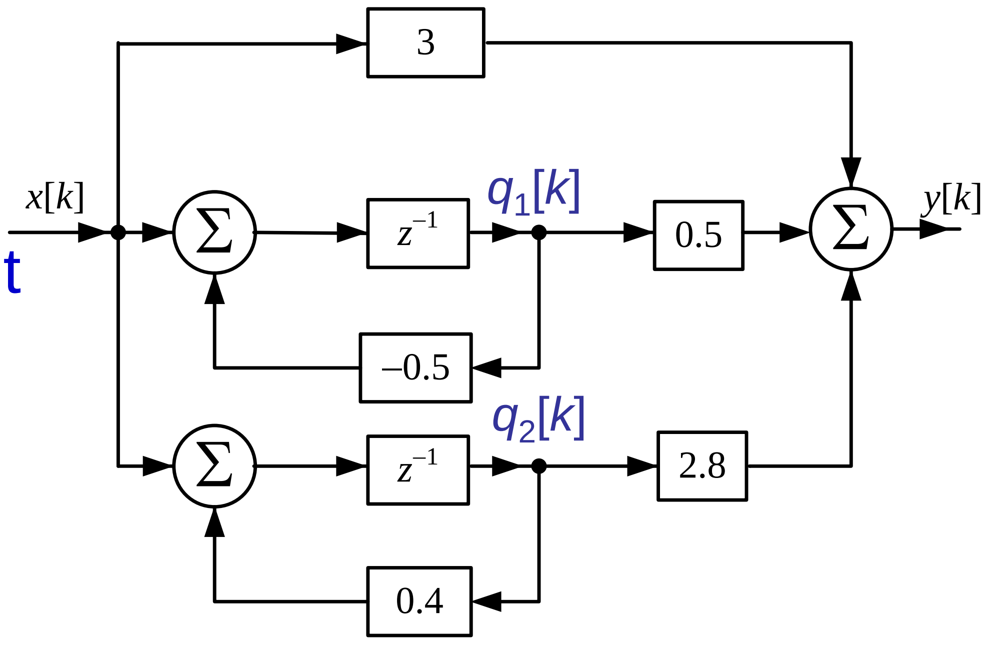
$$y[k] = \begin{bmatrix} 1.4 & -0.9 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + 3x[k]$$

State equations and output equations are established by system functions

(3) Parallel block diagram

$$H(z) = 3 + \frac{0.5z^{-1}}{1 + 0.5z^{-1}} + \frac{2.8z^{-1}}{1 - 0.4z^{-1}}$$

Select the output end of the delay as the state variables $q_1[k]$ and $q_2[k]$ of the system:



Equation of state and output equation:

$$q_1[k+1] = -0.5q_1[k] + x[k]$$

$$q_2[k+1] = 0.4q_2[k] + x[k]$$

$$y[k] = 0.5q_1[k] + 2.8q_2[k] + 3x[k]$$

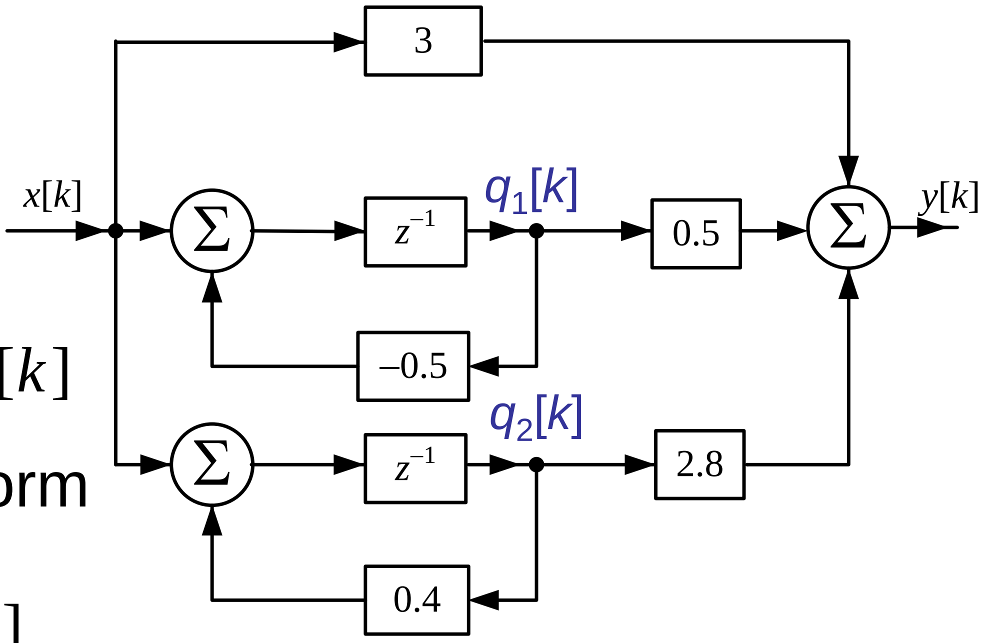
State equations and output equations are established by system functions

(3) Parallel block diagram

$$q_1[k+1] = -0.5q_1[k] + x[k]$$

$$q_2[k+1] = 0.4q_2[k] + x[k]$$

$$y[k] = 0.5q_1[k] + 2.8q_2[k] + 3x[k]$$



Output equation in matrix form

$$y[k] = \begin{bmatrix} 0.5 & 2.8 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + 3x[k]$$

State equation in matrix form

$$\begin{bmatrix} q_1[k+1] \\ q_2[k+1] \end{bmatrix} = \begin{bmatrix} -0.5 & 0 \\ 0 & 0.4 \end{bmatrix} \begin{bmatrix} q_1[k] \\ q_2[k] \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} x[k]$$

State equations and output equations are established by system functions

The simulation block diagram of discrete system is drawn by system function.

The state equation and output equation of the system are established by the simulation block diagram.

The equation of state and the output equation are different in the simulation block diagram of different structures.

Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you!